

Program : Diploma in Automation and Robotics	
Course Code : 4336	Course Title: Linear IC and Power Electronics Lab
Semester : 4	Credits: 1.5
Course Category: Engineering Science	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To design and implement application circuits using Analog ICs..
- To design and simulate the application circuits with Analog Integrated Circuits.
- To design and implement AC-DC rectifier circuits
- To design and implement DC to AC inverter circuits.
- To design and implement DC-DC converter circuits.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic concepts of Electronics circuit.		Analog and Digital Electronics Lab	3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Design and implement application circuits using Analog ICs.	15	Applying
CO2	Design and simulate the application circuits with Analog Integrated Circuits using simulation tools.	6	Applying
CO3	Familiarizing various power electronics switches and obtain their characteristics	6	Applying
CO4	Design and implementation of simple power electronics converters	12	Applying
	Lab Exam	6	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3						
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Design and implement application circuits using Analog ICs.		
M1.01	Familiarization of Operational amplifiers - Inverting and Non inverting amplifiers, frequency response	2	Applying
M1.02	Measurement of Op-Amp parameters.	1	Applying
M1.03	Integrator and Differentiator (With Frequency Response)	3	Applying
M1.04	Difference Amplifier and Instrumentation amplifier (3 op-amp configuration)	3	Applying
M1.05	RC Phase shift and Wien bridge oscillator using Op-Amp	3	Applying
M1.06	Astable and Monostable multivibrator using Timer IC NE555	3	Applying
CO2	Simulation experiments [The experiments shall be conducted using SPICE]		
M2.01	Simulation of Integrator and Differentiator circuits.	1	Applying
M2.02	Simulation of Precision rectifiers using Op-Amp	2	Applying
M2.03	Simulation of RC Phase shift and Wien bridge oscillator using Op-Amp	1	Applying

M2.04	Simulation of Astable and Monostable multivibrator using Timer IC NE555	2	Applying
	Lab Exam I	3	
CO3	Familiarizing various power electronics switches and obtain their characteristics		
M3.01	To study the V-I characteristics of SCR.	2	Applying
M3.02	To study the Latching & Holding current of SCR.	1	Applying
M3.03	To study the V-I characteristics of MOSFET.	2	Applying
M3.04	To study the V-I characteristics of TRIAC.	1	Applying
CO4	Design and implementation of simple power electronics converters		
M4.01	To study of single phase half wave controlled rectifier circuit with R & R-L load.	3	Applying
M4.02	To study of single phase full wave controlled bridge rectifier circuit with R & R-L load.	3	Applying
M4.03	Implementation of dc-dc buck converter	3	Applying
M4.04	Implementation of single phase half/ full bridge MOSFET/ IGBT inverter	3	Applying
	Lab Exam II	3	

Online Resources:

Sl No	Website Link
1	https://nptel.ac.in/courses/108/108/108108114
2	https://nptel.ac.in/courses/117/107/117107094
3	https://www.multisim.com/discover/
4	https://youtube.com/playlist?list=PLp6ek2hDcoND7i5-DAD9mPmYF1Wg6ROdO